PRACTICAL COURSE WORKBOOK

Wearables, IoT & Digital Health Devices

Work with digital-health information responsibly

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a safe, fictional health-data workflow and review
SKILLS	Wearables, IoT, BLE, Health APIs, Fitbit SDK, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Wearables appropriately in a realistic, bounded task.
- Use IoT appropriately in a realistic, bounded task.
- Use BLE appropriately in a realistic, bounded task.
- Use Health APIs appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners exploring health technology without replacing clinical expertise.

BEFORE YOU BEGIN

- Basic data literacy

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Wearables, IoT & Digital Health Devices is a structured, practical course covering Wearables, IoT, BLE, Health APIs, Fitbit SDK. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Wearables: Health context	1. Care pathways with Wearables 2. Health data with IoT 3. Stakeholders with BLE
02 IoT: Digital systems	1. Standards with Health APIs 2. Interoperability with Fitbit SDK 3. Workflows with Wearables
03 BLE: Quality and safety	1. Validation with IoT 2. Privacy with BLE 3. Human oversight with Health APIs
04 Health APIs: Applied evaluation	1. Implementation with Fitbit SDK 2. Monitoring with Wearables 3. Equity with IoT

HOW TO USE EACH LESSON PAGE

1. Read the objective and identify the decision it supports.
2. Attempt the retrieval prompt before checking the online explanation.
3. Reproduce the worked example with the stated input.
4. Test one normal case and one boundary or failure case.
5. Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Wearables: Health context

Apply Wearables through care pathways, health data, stakeholders.

LESSON 01 / 18 MIN

Care pathways with Wearables

Learning objectives

- Explain care pathways in the context of Wearables, IoT & Digital Health Devices.
- Apply Wearables to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms care pathways, Wearables, healthcare

Retrieval prompt In Wearables, IoT & Digital Health Devices, which evidence best supports a care pathways result produced with Wearables?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 02 / 22 MIN

Health data with IoT

Learning objectives

- Explain health data in the context of Wearables, IoT & Digital Health Devices.
- Apply IoT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms health data, IoT, healthcare

Retrieval prompt In Wearables, IoT & Digital Health Devices, which evidence best supports a health data result produced with IoT?

Evidence to keep [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 03 / 26 MIN

Stakeholders with BLE

Learning objectives

- Explain stakeholders in the context of Wearables, IoT & Digital Health Devices.
- Apply BLE to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms stakeholders, BLE, healthcare

Retrieval prompt	In Wearables, IoT & Digital Health Devices, which evidence best supports a stakeholders result produced with BLE?
Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

IoT: Digital systems

Apply IoT through standards, interoperability, workflows.

LESSON 04 / 30 MIN

Standards with Health APIs

Learning objectives

- Explain standards in the context of Wearables, IoT & Digital Health Devices.
- Apply Health APIs to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

standards, Health APIs, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a standards result produced with Health APIs?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 05 / 18 MIN

Interoperability with Fitbit SDK

Learning objectives

- Explain interoperability in the context of Wearables, IoT & Digital Health Devices.
- Apply Fitbit SDK to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

interoperability, Fitbit SDK, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a interoperability result produced with Fitbit SDK?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 06 / 22 MIN

Workflows with Wearables

Learning objectives

- Explain workflows in the context of Wearables, IoT & Digital Health Devices.
- Apply Wearables to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

workflows, Wearables, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a workflows result produced with Wearables?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

BLE: Quality and safety

Apply BLE through validation, privacy, human oversight.

LESSON 07 / 26 MIN

Validation with IoT

Learning objectives

- Explain validation in the context of Wearables, IoT & Digital Health Devices.
- Apply IoT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

validation, IoT, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a validation result produced with IoT?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 08 / 30 MIN

Privacy with BLE

Learning objectives

- Explain privacy in the context of Wearables, IoT & Digital Health Devices.
- Apply BLE to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

privacy, BLE, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a privacy result produced with BLE?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 09 / 18 MIN

Human oversight with Health APIs

Learning objectives

- Explain human oversight in the context of Wearables, IoT & Digital Health Devices.
- Apply Health APIs to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

human oversight, Health APIs, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a human oversight result produced with Health APIs?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Health APIs: Applied evaluation

Apply Health APIs through implementation, monitoring, equity.

LESSON 10 / 22 MIN

Implementation with Fitbit SDK

Learning objectives

- Explain implementation in the context of Wearables, IoT & Digital Health Devices.
- Apply Fitbit SDK to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

implementation, Fitbit SDK, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a implementation result produced with Fitbit SDK?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 11 / 26 MIN

Monitoring with Wearables

Learning objectives

- Explain monitoring in the context of Wearables, IoT & Digital Health Devices.
- Apply Wearables to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

monitoring, Wearables, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a monitoring result produced with Wearables?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 12 / 30 MIN

Equity with IoT

Learning objectives

- Explain equity in the context of Wearables, IoT & Digital Health Devices.
- Apply IoT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

equity, IoT, healthcare

Retrieval prompt

In Wearables, IoT & Digital Health Devices, which evidence best supports a equity result produced with IoT?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE
Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY
Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW
Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Wearables, IoT & Digital Health Devices project using Wearables, IoT, BLE.

DELIVERABLE	A working Wearables, IoT & Digital Health Devices artefact plus an evidence-based self-review.
TOOLS	A suitable Wearables environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Wearables.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This material is educational and does not provide medical advice or replace qualified clinical judgement.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE
Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE
What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION
What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION
Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Global strategy on digital health 2020-2027

World Health Organization - reviewed 2026-08-21

https://www.who.int/publications/i/item/9789240116870

FHIR specification

HL7 International - reviewed 2026-08-21

https://hl7.org/fhir/

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Wearables scope.